



Swallowing (Deglutition)

Physiology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. Name the three stages of swallowing, in order.
2. Which stage is voluntary, and which two are involuntary?
3. What two structures close off the nasal cavity and the airway during the pharyngeal stage?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Deglutition — the three stages

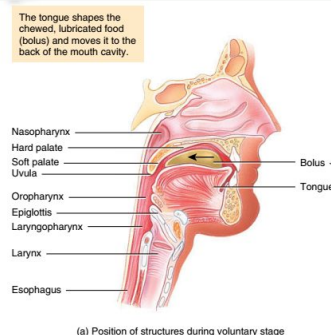
Swallowing moves food from the mouth into the stomach. It involves the mouth, pharynx and esophagus, is helped along by saliva and mucus, and always happens in the same three steps.

Stage	Voluntary / involuntary	Control centre
1. Voluntary	Voluntary	Cerebral cortex (initiation)
2. Pharyngeal	Involuntary	Deglutition centre — medulla oblongata & lower pons
3. Esophageal	Involuntary	Peristaltic reflex (myenteric plexus)

2 Stage 1 — The voluntary stage

Mechanism

The bolus is forced to the back of the oral cavity and into the **oropharynx** by the tongue moving **upward and backward** against the palate.

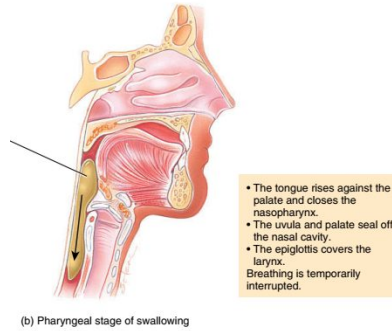


Position of structures during the voluntary stage — bolus pushed toward the oropharynx.

3 Stage 2 — The pharyngeal stage

NEVER MISS: this stage closes off **two** separate routes — the nasopharynx (soft palate + uvula) **and** the larynx (epiglottis) — protecting against both nasal reflux and aspiration.

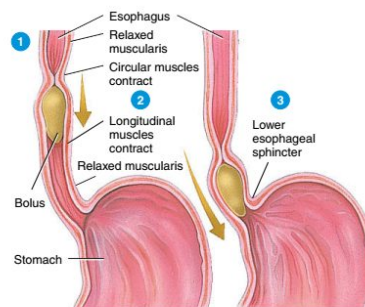
Trigger	Bolus stimulates receptors in the oropharynx → impulses to the deglutition centre (medulla oblongata & lower pons)
Nasopharynx closure	Returning impulses move the soft palate and uvula upward , sealing off the nasopharynx — prevents food/liquid entering the nasal cavity
Airway closure	The epiglottis closes off the opening to the larynx — prevents the bolus entering the respiratory tract
Transit	Bolus moves through the oropharynx and laryngopharynx; the upper esophageal sphincter relaxes and the bolus enters the esophagus



Pharyngeal stage — soft palate and uvula seal the nasopharynx; epiglottis covers the larynx.

4 Stage 3 — The esophageal stage

Mechanism	Peristalsis — coordinated contraction and relaxation of the circular and longitudinal muscularis layers
Step 1	Circular muscle fibres above the bolus contract → constrict the wall → squeeze the bolus toward the stomach
Step 2	Longitudinal fibres below the bolus contract → shorten that segment, widening it to receive the bolus. Steps 1–2 repeat in waves.
Step 3	Once the bolus reaches the lower esophageal sphincter , it relaxes and the bolus enters the stomach



Esophageal peristalsis: circular muscle contracts above the bolus, longitudinal muscle contracts below it.

5 Timing & lubrication

Bolus type	Transit time
Solid / semisolid	4–8 seconds
Very soft food / liquid	~1 second

Mucus secreted by **esophageal glands** — not saliva — lubricates the bolus through the esophagus and reduces friction.

P Must-remember pearls

- The pharyngeal stage closes **two** routes: nasopharynx (soft palate/uvula) and larynx (epiglottis).
- The deglutition centre sits in the **medulla oblongata and lower pons**.
- Circular muscle contracts **above** the bolus; longitudinal muscle contracts **below** it — the peristaltic principle repeats down to the LES.
- Liquids transit in ~1 second; solids take 4–8 seconds.
- **Esophageal gland** mucus lubricates the bolus once it's past the pharynx — not saliva.

Q Test yourself

1. A stroke damaging the medulla oblongata would most directly impair which stage of swallowing?

- A. Voluntary stage
- B. Pharyngeal stage
- C. Esophageal stage
- D. All three equally

2. Closure of the nasopharynx during swallowing is achieved by:

- A. Epiglottis folding over the larynx
- B. Soft palate and uvula moving upward
- C. Upper esophageal sphincter contracting
- D. Tongue pressing against the palate

3. During esophageal peristalsis, which muscle layer contracts immediately ABOVE the bolus?

- A. Longitudinal muscle
- B. Circular muscle
- C. Both simultaneously
- D. Neither — it's purely gravity-driven

4. A patient who aspirates liquid into the airway while swallowing most likely has dysfunction of:

- A. The upper esophageal sphincter
- B. Epiglottis closure over the larynx
- C. The lower esophageal sphincter
- D. Tongue elevation against the palate

5. Relaxation of which structure finally allows the bolus to enter the stomach?

- A. Upper esophageal sphincter
- B. Pyloric sphincter
- C. Lower esophageal sphincter
- D. Ileocecal valve

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 — **B** — the deglutition centre that runs the pharyngeal stage sits in the medulla oblongata and lower pons.
- 2 — **B** — the soft palate and uvula move up to seal the nasopharynx.
- 3 — **B** — circular muscle above the bolus contracts to squeeze it downward; longitudinal muscle below relaxes/contracts to receive it.
- 4 — **B** — failure of epiglottis closure lets the bolus enter the respiratory tract instead of the esophagus.
- 5 — **C** — the lower esophageal sphincter relaxes to let the bolus pass into the stomach.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Physiology GIT Block · MEDucation by Mattin Majid · GIT Module slides, 2025-2026
Study alongside the lecture, not instead of it.